

A VR Enabled Visualization System for Race Suit Design

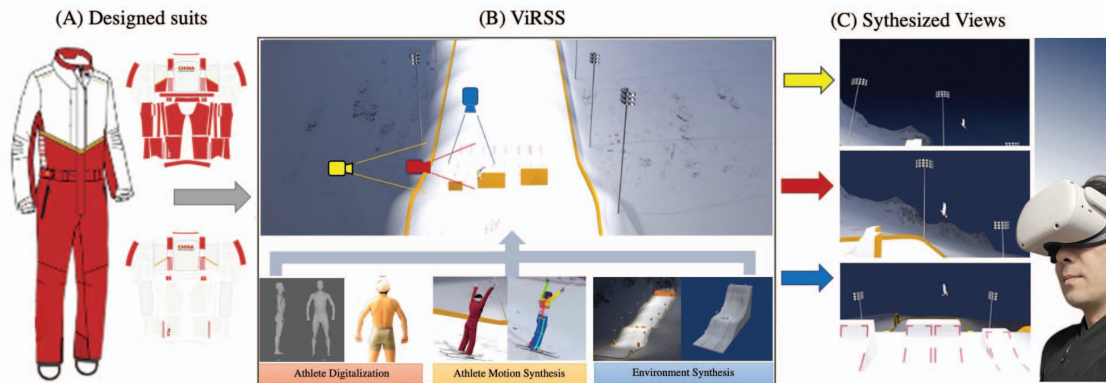
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Figure 1: The overview of our approach. (a) The input is the designed race suit. (b) Our system, *ViRSS*, synthesizes the scenario of the real application, utilizing the constructed assets, e.g., digitalized athlete, motion, and environment. (c) Through a VR helmet, the designer may observe the design from any view to facilitate further improvement.

ABSTRACT

In this paper, we propose a virtual reality technique enabled visualization system for race suits design, allowing designers to observe the design work from a near-real perspective of users. We take the skiing-jumping race suit design as an example to demonstrate the core idea. Firstly, the system automatically constructs a virtual competition venue according to the real field settings. A digital avatar scanned from an athlete is dressed in a designed race suit. Then the avatar is given a sequence of preset sports motions to act in the virtual competition venue. The system can synthesize the traditional user's views for designers to observe via virtual reality devices and improve the efficiency of the race suit design workflow.

Keywords: Design Workflow, CAD, Digital Clothing.

1 INTRODUCTION

In competitive sports, race suit is one of the most critical considerations in the final preparation for athletes, which may affect athletes' performance and safety. From the perspective of a race suit designer, the design process is quite different from simple sportswear design, leading to some challenging problems during the design workflow. One is the difficulty of obtaining feedback about the appearance.

In the traditional workflow, a designer iteratively polishes his work. That is, designing, making samples, trying on, and returning to the first stage for adjustment according to the appearance feedback. However, this workflow is challenging to implement in the race suit design because the athlete has less time to try on the samples during the intensive training. Another challenge arises from the characteristics of race suits. The athlete is supposed to perform specific sports motions in a certain environment. When the conditions change, e.g., motion style, background, lighting condition, or observation views, the effect of the race suit may also change. The designer needs a tool to evaluate the design effect in all environmental contexts.

To address the above challenges, this paper proposes a Virtual reality techniques based race Suits System, *ViRSS*, leveraging virtual

reality techniques to optimize the design process by constructing an immersive evaluation system for designers. Firstly, we build digital assets and environments through digitalization for the evaluation system, including scanned athlete's body shapes, parameterized race suits, and digitalized competition environment. Then, with such digital assets, *ViRSS* constructs a virtual environment, in which the avatar of the athlete, dressed up with the designed race suit, is placed into the virtual competition venue to perform desired sports motions. *ViRSS* synthesizes users' views, and the designer could use VR devices, e.g., Oculus, to observe the wearing effect under the near-real environment context immersively.

Current researches mainly focus on improving clothing simulation, such as enabling the simultaneous animation of thousands of detailed garments in real-time [5], endowing individuals with realism and variety [1], and animating realistic clothing on synthetic bodies of any shape and pose without manual intervention [2]. However, the wearing context is ignored. For some specific types of garment, the context information is essential, e.g., the race suits.

Thus, in this paper, we attempt to design a system for synthesizing the wearing context for designers. Compared with the traditional suit design method, the proposed *ViRSS* enables race suit designers to evaluate the design work under various contexts in an immersive way with lower cost.

The main contributions of this paper are summarized as follows:

- (1) Propose a new problem of evaluating the appearance of the race suit design, considering the wearing context;
- (2) Devise a pipeline for imitating the application scenario of the race suits in a VR environment with various wearing context configurations to facilitate the race suit design.

2 METHOD

We propose a computer-aided system based on VR techniques for race suit design to supplement the traditional design workflow. The pipeline of *ViRSS* is shown in Fig. 1, comprising two components: (a) digitalization and (b) simulation. The digitalization component aims to construct the digital assets used in the subsequent simulation, including digital competition venues, athletes, and sports motions. The simulation component uses the constructed digital assets to simulate the appearance of a design work according to the desired configurations. The configurations of the simulation can be customized by the designers or sampled randomly. *ViRSS* provides three configurations: lighting condition, sports motion, and observa-

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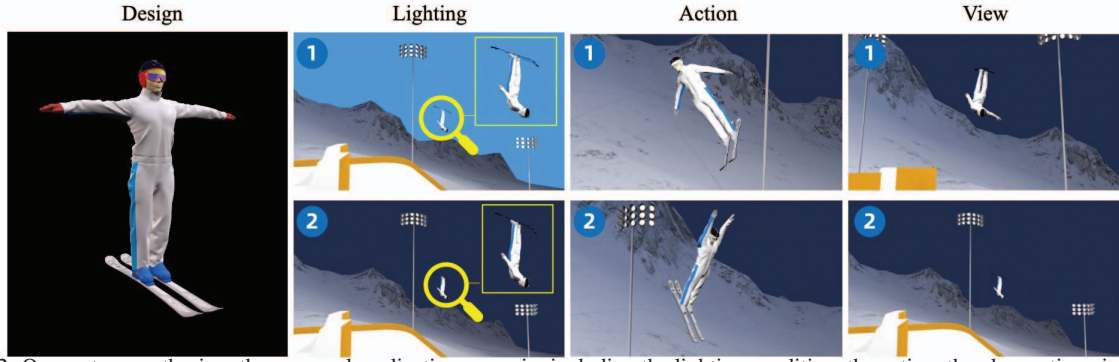


Figure 2: Our system synthesizes the near-real application scenario, including the lighting condition, the action, the observation view, and the field, to facilitate the appearance evaluation of the designed race suits. The designer may customize specific configurations, as shown above.

tion view. Then, the designer may observe the athlete's appearance from the synthesized view, as shown in Fig. 1 (c).

2.1 Athlete Digitalization

In *ViRSS*, the digital athlete is placed in a virtual competition venue, performing some preset sports motions to demonstrate the appearance for the designer to evaluate the result. Thus, we scan the 3D body shape of a specific athlete through a handheld scanner, which is used as a virtual model to try on the digital samples.

A scanned model usually accompanies noises. We process the original scanned model by estimating a parametric body model instead of the original scanned result. We utilize SMPL [3] as the parametric representation. The avatar is represented as a function $M(\theta, \beta)$, with pose parameters θ and shape parameters β returning a body mesh $M \in \mathbb{R}^{N \times 3}$. Then we apply [3] approach to estimate the parameters of the body shape. The core idea is to learn a regressor through a neural network, mapping a 3D scanned body shape to the 3D body shape parameter space. The obtained parameters are utilized in subsequent appearance simulations.

2.2 Athlete Motion Synthesis

Performing sports motions with high quality is the most important task. An excellent design may guarantee the necessary functionality and provide an attractive appearance when the athlete performs the desired actions. To reflect the appearance as real as possible, we employ computer vision techniques to establish a motion dataset comprising 50 sports videos and the corresponding pose parameters. We collect 50 videos; each video is a clip of a traditional sports motion. Then we use [4] to estimate the pose parameters for each frame, yielding a sequence of sports motion pose in the way of 3D skeleton. The sequence is written as $\Theta = \{x^t, y^t, z^t, \theta_i^t, \phi_i^t, \psi_i^t, i \in [1, 14]\}$, where (x_t, y_t, z_t) representing the location of the athlete at time t , $(\theta_i^t, \phi_i^t, \psi_i^t)$ representing the Euler angles of the i th joint.

In the simulation phase, we re-target the sports motion on the avatar to rig the dressed avatar. Then, we combine the global and local motion to synthesize the sports motion in the virtual environment. After that, the designer can view the appearance under different motions.

2.3 Competition Venues Environment Synthesis

The competition venue usually is built based on standard specifications, e.g., the size, decent, materials of the surface of the platform, ensuring the match field condition is stable. So we construct the virtual field referring to the public standard of the competition venues.

To imitate the real scene in which the race suits apply as much as possible, we also match the common background manually according to the functionality type. For example, ski mountaineering usually is on hills; speed skating is in an indoor stadium; ski jumping takes place outdoor field. Based on the background, we place the necessary components of the field. Take the ski jumping program as an example. We set the ski ramp following the guidance of the International Ski Federation (FIS), applying traditional materials. Furthermore, we preset common lighting conditions for designers

to select so that the designer can evaluate the design work under various types of illumination, e.g., the result at night compared with the one in the day.

2.4 View Synthesis

ViRSS provides various views for the designer to observe the appearance of the sample race suit. The view synthesis is implemented by placing the camera in the virtual scene. Although we can synthesize the scene in the virtual scene under any view, considering the settings in the real competition venue, we design some common observation views for the designers, including the view for the umpire, audience, and broadcasting. The designer can observe the virtual garment's appearance immersively through a VR device.

We apply *ViRSS* to synthesize the appearance of race suit designs. Each visualization condition includes lighting conditions, observation views, and sports motions. The system makes the evaluation by amateur users possible because it could provide a friendly way to observe the design. So the designer may collect more feedback painlessly to improve the design. That is the primary motivation for the design of *ViRSS*. A more comprehensive perceptual study will be an exciting avenue for future research.

3 CONCLUSION

This paper proposes *ViRSS* to help designers evaluate design work under various conditions. Through our system, designers may evaluate their designs more comprehensively. The experiment results support that designers change their impression of the same design via different observation conditions.

Although *ViRSS* is proposed for improving the workflow efficiency for race suit design, it sheds light upon applications such as visualization and evaluation for design work. For example, we may apply our system to broader design fields, which usually need to consider the environmental context of using the design, such as stage design, exhibition design, group show, and so on.

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